

DENVar: A Global Dynamic Estimate of NDVI Variance

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Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and projects. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance around the mean (i.e., stochasticity) has gained more attention, particularly as climate change and human-induced rapid environmental change transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean and the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. We show that DENVar correlates nonlinearly with **XXX** and can be used to test many hypotheses related to environmental stochasticity, such as optimal foraging theory and regime shifts as well as for conservation. We provide a publicly and freely available repository for downloading rasters of daily and long-term averages of DENVar and the related estimates of mean NDVI.

1 Introduction

The Normalized Difference Vegetation Index (NDVI) is a proxy of primary productivity used in a wide range of disciplines, including ecology [6, 7], landscape phenology [8, 9], agriculture [10], forestry [11], and conservation [12, 13]. High-resolution, remotely-sensed NDVI data have allowed analysts to estimate plant biomass without fieldwork [14, 15] and leverage such estimates for studying large-scale plant dynamics [6] and their effects on animal movement behavior [8, 16]. Still, most NDVI work is limited to quantifying changes in and responses to mean NDVI while ignoring changes in the variance around the mean.

Environmental stochasticity has been studied for decades [17–20], but recent work has drawn more attention to the effects environmental stochasticity has at a wide range of scales, from individual organisms [21, 22] to landscapes [13, 23, 24]. However, estimating trends in NDVI variance requires more nuanced modeling than estimating changes in mean NDVI, particularly when environments are temporally dynamic and spatially heterogeneous. The scale at which processes are observed determine how individuals’ perception of and ability to respond to change [25]. A forest fire is severely disruptive and deadly for many insects [26], while birds and large mammal may not struggle to escape the fire and or establish a new range [27, 28]. Similarly, the effects of the fire recovery will depend on its duration relative to other organisms’ phenology and lifespan [29]. Therefore, distinguishing between a change in mean NDVI and a stochastic fluctuation requires a conscious decision about the scale at which to consider the data.

Previous work suggests that environmental stochasticity can reduce limit species’ ability to adapt or specialize [30], alter species’ movement behavior [21], and even affect extinction risk at both the population and species levels [19, 31, 32]. Still, the effects of environmental stochasticity on species dynamics and ecosystem stability remain understudied [33]. Global, temporally dynamic estimates of NDVI variance would facilitate research by building upon the widespread use and interpretability of NDVI while also removing the barriers of computational demand and statistical knowledge that are required for quantifying environmental stochasticity. However, since NDVI is not a direct measurement primary productivity but a proxy, it should be interpreted carefully [7]. NDVI is prone to sensitivity loss in very green environments [34, 35], values vary across sensors [36–38], and the coarse spatiotemporal scale of satellite-derived NDVI can be a substantial barrier when assessing fine-scale trends [39, 40]. However, careful statistical modeling can help overcome some of these limitations. In particular, Generalized Additive Models (GAMs) are particularly convenient due to their ability to learn flexibly from data [41, 42] and handle very large spatiotemporal datasets [43, 44].

We present a new measure of environmental stochasticity, which we name the Dynamic Estimate of NDVI Variance (DENVar). We calculate DENVar using 44 years of NDVI data from the AVHRR and VIIRS collection of sensors and GAMs. Although satellites with AVHRR and VIIRS sensors provide data at a coarser spatial resolution (0.05°) than other satellites (e.g., 250m for MODIS), they are the only sensors to provide data at a daily frequency and for more than 40 years. Our use of GAMs provides a flexible yet interpretable approach for estimating trends in mean and variance in NDVI without the rigidity of other more traditional methods (e.g., ARIMA) or the opaqueness of

machine learning models such as neural networks [41, 42]. We illustrate the relationships between DENVar and multiple, relevant ecological metrics, and we provide access to a free repository of daily, global estimates of DENVar and the corresponding estimates of mean NDVI.

2 Materials and Methods

2.1 Choice of color palettes

We represent NDVI using a modified version of Fabio Crameri’s divergent *bukavu* palette [1, 45], which has high-contrast for deuteranope and protanope vision. For NDVI values between -0.1 and 1, the colors also have sufficient contrast for the colors to also be distinguishable by strong tritanope and achromatic vision. Fig. A1 contains an approximate representation of how each vision type would perceive each color palettes used. We obtained all color palettes with three or more colors from the *khroma* package (v. 1.14.0, [4]) for R (v. 4.4.1, [46]).

2.2 Experimental and Technical Design

- include a diagram to show the entire experimental design and illustrate the most significant elements: materials, treatments, measurements, data collection, methods of data analysis
- This will facilitate the editors and reviewers to understand and follow the whole concept, design, and results.

We downloaded NDVI image composites directly from the servers for data from AVHRR sensors (1981-06-24 to 2013-12-31; <https://ladsweb.modaps.eosdis.nasa.gov/archive/allData/>; [47]) and VIIRS sensors (2014-01-01 to 2025-06-29; <https://ncei.noaa.gov/data/>; [48]). Fig. A2 shows the availability of rasters over time for both sets of sensors. We removed all pixels labeled as "cloudy" for AVHRR rasters and "probably cloudy" or "confidently cloudy" for VIIRS rasters. Additionally, we removed pixels with erroneously high NDVI values at high latitudes between September and March using latitudinal cutoffs based on the monthly empirical cumulative distribution function of NDVI across latitude (Fig. B2). We did not remove values at latitudes $< 60^{\circ}\text{S}$ since we fully excluded Antarctica from the analyses.

Elevation above sea level was obtained using the `get_elev_raster()` function from the *elevatr* package for R (v. 0.99.0, [49]) with an original resolution of 0.076° (Fig. A6). To remove any biases towards negative NDVI values due to water, we removed all raster cells with $> 40\%$ water cover. We estimated the proportion of water in each cell by rasterizing a shapefile of main water bodies at a resolution of 0.05° (Fig. A7).

To ensure the number of rows for each dataset was below the maximum data frame size in R ($3.17 \times 10^{10} > 2^{31} - 1 \approx 0.21 \times 10^{10}$), we reduced the dataset size by splitting the dataset across four spatial groups (Fig. A4) and spatiotemporally averaged the (cleaned) rasters by a factor of four. The final dataset thus had one raster every four days with a spatial resolution of $0.05^{\circ} \times 4 = 0.2^{\circ}$. Despite the substantial reduction in dataset size, the dataset creation and modeling were still challenging

for the machine used, which had 2.2 TB of RAM and two Intel Xeon Platinum 8462Y+ processors (64 cores total).

2.3 Statistical Analysis

We estimated spatiotemporal trends in mean NDVI using Generalized Additive Models (GAMs) via the `mgcv` package for R (v. 1.9-3, [41]). To reduce modeling fitting times with negligible losses to model accuracy, we used the `bam()` function with fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization (`discrete = TRUE`; [43, 44]). Ideally, NDVI should be modeled using beta location-scale models (after the linear transformation $Y^* = \frac{Y+1}{2}$) to account for the fact that: (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent on the mean (and vice-versa), since ecosystems with very low NDVI (e.g., rock, ice, or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in NDVI. However, preliminary tests showed that fitting times for beta models were prohibitive, especially for beta location-scale models. In contrast, Gaussian models fit substantially faster and provided very similar spatialtemporal estimates of mean NDVI (see Appendix B). For each region shown in Fig. A4, the model for the mean NDVI had the structure below:

$$\begin{cases} \text{NDVI} \sim \text{Normal}(\mu, \sigma^2) \\ \mu = \beta_0 + f_1(\text{prop_water}) + f_2(\text{elev_m}) + f_3(\text{doy}) + f_4(\text{year}) + f_5(\text{lat, long}) + f_{6T}(\text{doy, year}) + f_{7T}(\text{doy, lat, long}) + f_{8T}(\text{year, lat, long}) \end{cases}, \quad (1)$$

where, for a given (aggregated) pixel, `prop_water` is the proportion of water within the pixel, `elev_m` is elevation above sea level in meters, `doy` is integer day of year (1-366), `year` is integer year (1981-2025), and `lat` and `long` are latitude and longitude, respectively. $f_j(\cdot)$ indicate the j^{th} marginal smooth functions, while $f_{jT}(\cdot)$ indicate the j^{th} tensor product interaction terms. Latitude and longitude were modeled using a spline-on-the-sphere basis (`bs = 'sos'`; [41]), while seasonal terms used cyclical cubic splines (`bs = 'cc'`) with edge knots 0.5 and 366.5.

HERE

A second GAM estimated the variance in NDVI around the mean estimated by the model above. The model had an identical structure to the model for the mean, with the exception that (1) the response variable was the squared residual from the first model, such that the model estimated the mean squared residual (i.e., the variance) for a given point in time and space, and (2) a smooth term of the estimated mean NDVI. *Estimating the variance as the spatially or spatiotemporally varying mean squared residuals included the pixel-level squared bias.* The model is available below:

formula

3 Results

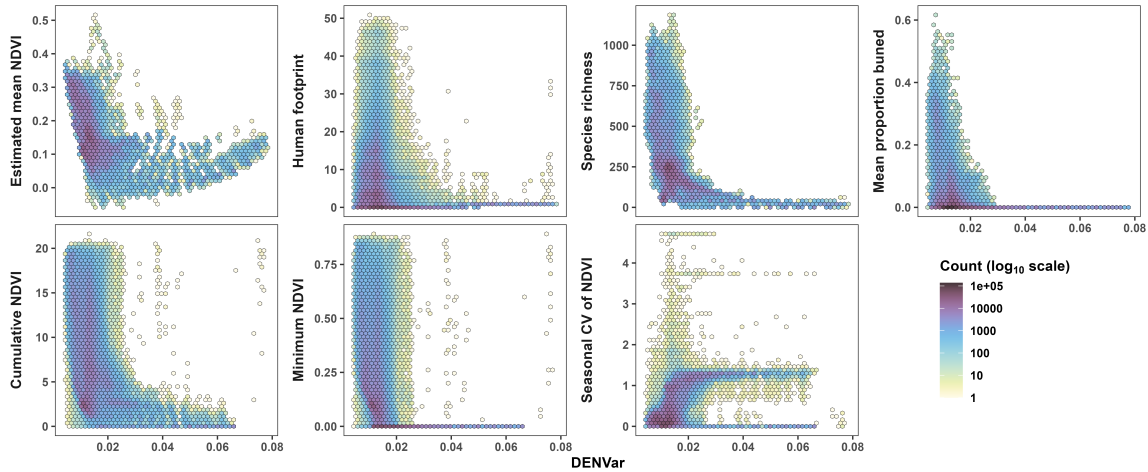
variance is likely overestimated in areas near sharp changes in NDVI, but this depends on scale

Figures:

- two orthographic projections of DENVar: northern and southern hemispheres on spring solstice
OR Robinson projection of the world with biomes to see if fill matches biome boundaries
- plot grid of mean over space, `s(year)`, `s(doy)`, `s(elev)`
- plot grid of DENVar over space, `s(year)`, `s(doy)`, `s(elev)`, `s(mean NDVI)`
- hex plots of pred vs obs for mean and var
- hex plots of DENVar with other variables (add temperature, annual precip, extreme weather event, fires, invasive species, max body size, disease occurrence, ...)
- supplementary material: histograms of mean around flat map of biomes (already created, but change colors?)

3.1 Correlations with other metrics

- temperature
- seasonal temperature range
- precip
- seasonal temperature range
- spp diversity
- gross primary productivity
- max animal weight
- <https://silvis.forest.wisc.edu/globalwui/>
- <https://silvis.forest.wisc.edu/data/dhis/>: NDVI16 (Normalized Difference Vegetation Index), EVI16 (Enhanced Vegetation Index), FPAR8 (Fraction absorbed Photosynthetically Active Radiation), LAI8 (Leaf Area Index), GPP8 (Gross Primary Productivity)



3.2 Obtaining the data

- Data should be available as daily rasters on a github repo?
- make an app for data annotation?

4 Discussion

We have presented a new spatiotemporally dynamic metric of environmental stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DENVar. In this section, we discuss the interpretation of DENVar in greater depth, and we argue the advantages and limitations of the statistical methods used. We conclude by suggesting some use cases of DENVar within animal movement ecology, landscape ecology, and phenology, but we invite readers to consider how DENVar may be of use in their field and adjacent disciplines.

4.1 Interpreting DENVar: the importance of scale

difficulties in estimating variance

what counts as a change? predictable vs stochastic changes

smoothness of mean impacts smoothness and size of variance: depends on scale of interest (e.g., animals' ability to respond to, learn, and predict changes in the mean). Changes not attributed to trends in the mean are stochastic

DENVar is a good metric as long as the spatiotemporal scale of interest is similar or larger to that of DENVar. May not work well for small-scale processes such as the movement dynamics of a small rodent, the phenology of a small orchard when spatiotemporal heterogeneity is finer than DENVar’s resolution

4.2 Environmental stochasticity and animal movement ecology

The field of animal movement ecology has long shown that animals alter their movement in response to resource abundance or availability [50–53]. More recently, movement ecologists have started focusing more on resource stochasticity [18, 22, 33], but it is worth distinguishing between resource stochasticity and seasonality [54]. Seasonality is generally best interpreted as a cyclical change in the mean that many animals can learn to predict, as in the case of green-wave surfing [55, 56]. In contrast, environmental stochasticity refers to unpredictable changes, such as forest fires [ref](#), sudden snow storms [57], and unusual phenology [ref](#). Animals also respond to stochastic changes [21], but their ability to adapt depends on behavioral plasticity [58, 59], and memory [60–64].

evolutionary tipping points [65]

multifarious change [66]

extreme events, black swans [67, 68]

- migration routes are key habitat (Ortega et al., in prep), especially in stochastic environments

- can fail to adapt to changes in conditions [69]

- <https://www.sciencedirect.com/science/article/pii/S0960982220308484>

- <https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.15169>

- industrial development affects animals’ ability to respond to both predictable and unpredictable change (<https://www.nature.com/articles/s41559-022-01887-9>)

- effects of sudden deep snow in red desert on proghorn (<https://www.sciencedirect.com/science/article/abs/pii/S0960982220308484>)

- <https://www.nature.com/articles/s41467-023-37750-z>

- <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.4238>

- detecting regime shifts (in lakes: [70])

4.3 Environmental stochasticity and population dynamics

read lande_risks_1993

Environmental stochasticity reduces habitats' energetic balance and carrying capacity [30] as individuals struggle to rely on unpredictable resources and increase the instability of populations and communities [19]. Additionally, stochastic environments reduce the ability of specialist species to adapt and evolve while selecting for more generalist species [17]. Consequently, stochastic environments tend to have lower species diversity [13].

Recent work by Devries has suggested that NDVI stochasticity is an important metric for evaluating forest recovery after large fires, while Collinson has suggested that variance in NDVI is an important predictor of the occurrence of lightning-induced forest fires.

- jonzen_rapid_2006
- effects can be compounded by cumulative effects (e.g., loss of species diversity; [71])
- climate change affects plant phenology, but changes in situ may not match experimental estimates [72]
- reproduction of painted lady butterfly [73]
- wessling_seasonal_2018

213 4.4 Environmental stochasticity and fire ecology

214 4.5 Environmental stochasticity and phenology

215 5 Other useful references

216 • **READ**: Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated
217 Using Simulated Ecological Data (<https://doi.org/10.1371/journal.pone.0041010>)

218 • **READ**: <https://doi.org/10.1029/2024GL109298>

219 • **READ**: Site fidelity as a maladaptive behavior in the Anthropocene: <https://esajournals.onlinelibrary.wiley.com/doi>

220 • **READ**: Pettorelli N., Vik J.O., Mysterud A., Gaillard J.-M., Tucker C.J. & Stenseth N.Chr.
221 (2005). Using the satellite-derived NDVI to assess ecological responses to environmental
222 change. Trends in Ecology & Evolution 20, 503–510. <https://doi.org/10.1016/j.tree.2005.05.011>

223 • increase in variance before eutrophication in lakes: <https://doi.org/10.1002/lno.10355>

224 • uses ndvi variance: <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348>

225 • bro-jorgensen_using_2008: in the dry season preceding the rut topi antelope (*Damaliscus*
226 *lunatus*) respond to forage availability at the large scale but not the fine scale

227 • keith_predicting_2008

228 • chevin_adaptation_2010

229 • rickbeil_plasticity_2019

230 • mueller_search_2008

231 • pettorelli_using_2005

232 • nilsen_can_2005

233 • merkle_large_2016

234 • tian_evaluating_2015

235 • huang_commentary_2021

- 236 • fan_global_2016
- 237 • wang_stochastic_2019
- 238 • pease_ecological_2024
- 239 • xu_plasticity_2021
- 240 • calibrating across sensors: <https://ieeexplore.ieee.org/abstract/document/6809983>

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244 General

245 Thank others for any contributions, whether it be direct technical help or indirect assistance

246 Author Contributions

247 Describe contributions of each author to the paper, using the first initial and full last name.

248 Examples:

249 “S. Zhang conceived the idea and designed the experiments.”

250 “E. F. Mustermann and J. F. Smith conducted the experiments.”

251 “All authors contributed equally to the writing of the manuscript.”

252 Funding

253 Name financially supporting bodies (written out in full), followed by the funding awardee and asso-
 254 ciated grant numbers (if applicable) in square brackets.

255 Example:

256 “This work was supported by the Engineering and Physical Sciences Research Council grant
 257 numbers xxxx, yyyy; the National Science Foundation grant number zzzz; and a Leverhulme Trust
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260 of the authors, please name this employer. If the funder was involved in the manuscript writing,
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272 analysis of the work, the preparation or editing of the manuscript, or the decision to publish must
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282 access and why.

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Please group supplementary materials in the following order: materials and methods, figures, tables, and other files (such as movies, data, interactive images, or database files).

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A link to access the supplementary materials will be provided in the published article.

Supplementary Materials may include additional author notes—for example, a list of group authors.

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